28/10/2024 13:03:15

Compare Results

Old File:

USB_PD_R3_2 V1.0 2023-10_Part23.pdf

15 pages (651 KB)
15/04/2024 10:52:57

versus

New File:

USB_PD_R3_2 V1.1 2024-10_Appendix C VDM Examples.pdf

15 pages (309 KB)
28/10/2024 11:27:52

Total Changes

370

Text only comparison

Content

204 Replacements

90 Insertions

76 Deletions

Styling and Annotations

0 Styling

0 Annotations

Go to First Change (page 1)

C VDM Command Examples

C.1 Discover Identity Example

C.1.1 Discover Identity Command request

Table C.1, "Discover Identity Command request from Initiator Example" below shows the contents of the key fields in the *Message Header* and *VDM Header* for an *Initiator* sending a ***Discover Identity*** *Command* request.

Table C.1 Discover Identity Command request from Initiator Example

Bit(s)	Field	Value
	Message Header	
15	***Reserved***	0
14…12	***Number of Data Objects***	1 (*VDM Header*)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
	VDM Header	
B31…16	***Standard or Vendor ID (SVID)***	0xFF00 (***PD SID***)
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	000b
B7…6	***Command Type***	00b (*Initiator*)
B5	***Reserved***	0
B4…0	***Command***	1 (***Discover Identity***)

C.1.2 Discover Identity Command response - Active Cable.

Table C.2, "Discover Identity Command response from Active Cable Responder Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for a *Responder* returning *VDO*s in response to a ***Discover Identity*** *Command* request. In this illustration, the *Responder* is an active Gen2 cable which supports *Modal Operation*.

Table C.2 Discover Identity Command response from Active Cable Responder Example

Bit(s)	Field	Value
Message Header		
15	***Reserved***	0
14…12	***Number of Data Objects***	5 (*VDM Header* + *ID Header VDO* + *Cert Stat VDO* + *Product VDO* + *Cable VDO*)
11…9	***MessageID***	0…7
8	***Cable Plug***	1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
VDM Header		
B31…16	***Standard or Vendor ID (SVID)***	0xFF00 (***PD SID***)
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	000b
B7…6	***Command Type***	01b (*Responder ACK*)
B5	***Reserved***	0
B4…0	***Command***	2 (***Discover Identity***)
ID Header VDO		
B31	***USB Communications Capable as USB Host***	0 (not *USB Communications* capable as a *USB Host*)
B30	***USB Communications Capable as a USB Device***	0 (not data capable as a Device)
B29…27	***SOP' Product Type (Cable Plug/VPD)***	100b (*Active Cable*)
B26	***Modal Operation Supported***	1 (supports Modes)
B25…16	***Reserved***. ***Shall*** be set to zero.	0
B15…0	16-bit unsigned integer. ***USB Vendor ID***	USB-IF assigned *VID* for this cable vendor
Cert Stat VDO		
B31…0	32-bit unsigned integer	USB-IF assigned XID for this cable
Product VDO		
B31…16	16-bit unsigned integer. USB Product ID	Product ID assigned by the cable vendor
B15…0	16-bit unsigned integer. bcdDevice	Device version assigned by the cable vendor
Cable VDO1 (returned for Product Type "Active Cable")		
B31…28	***HW Version***	Cable HW version number (vendor defined)
B27…24	***Firmware Version***	Cable FW version number (vendor defined)
B23…21	***VDO Version***	010b (Version 1.2)

Table C.2 Discover Identity Command response from Active Cable Responder Example

Bit(s)	Field	Value
B20	***Reserved***	0
B19…18	***USB Type-C plug to USB Type-C/Captive***	10b (*USB Type-C®*)
B17	***EPR Capable (Active Cable)***	0 (not *EPR Capable*)
B16…13	***Cable Latency***	0001b (<10ns (~1m))
B12…11	***Cable Termination Type (Active Cable)***	11b (Both ends Active, *VCONN* required)
B10…9	***Maximum VBUS Voltage (Active Cable)***	00b (20V)
B8	***SBU Supported***	0 (SBUs connections supported)
B7	***SBU Type***	0 (SBU is passive)
B6…5	***VBUS Current Handling Capability (Active Cable)***	01b (3A)
B4	***VBUS Through Cable***	1 (Yes)
B3	***SOP'' Controller Present***	1 (*SOP''* controller present)
B2…0	***Reserved***	0
Cable VDO2 (returned for Product Type "Active Cable")		
B31…24	***Maximum Operating Temperature***	70
B23…16	***Shutdown Temperature***	80
B15	***Reserved***	0
B14…12	***U3/CLd Power***	010b (1-5mW)
B11	***U3 to U0 transition mode***	00b (U3 to U0 direct)
B10	***Physical connection***	0 (Copper)
B9	***Active element***	0 (Active Re-driver)
B8	***USB4 Supported***	0 (USB4 Supported)
B7…6	***USB 2.0 Hub Hops Consumed***	2
B5	***USB 2.0 Supported***	0 (*[USB 2.0]* supported)
B4	***USB 3.2 Supported***	0 (*[USB 3.2]* SuperSpeed supported)
B3	***USB Lanes Supported***	1b (Two lanes)
B2	***Optically Isolated Active Cable***	0 (Not supported)
B1	***USB4 Asymmetric Mode Supported***	0 (Not Supported)
B0	***USB Gen***	1b (Gen 2 or higher)

C.1.3 Discover Identity Command response - Hub.

Table C.3, "Discover Identity Command response from Hub Responder Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for a *Responder* returning VDOs in response to a ***Discover SVIDs*** *Command* request. In this illustration, the *Responder* is a *Hub*.

Table C.3 Discover Identity Command response from Hub Responder Example

Bit(s)	Field	Value
	Message Header	
15	***Reserved***	0
14…12	***Number of Data Objects***	4 (*VDM Header* + *ID Header VDO* + *Cert Stat VDO* + *Product VDO*)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
	VDM Header	
B31…16	***Standard or Vendor ID (SVID)***	0xFF00 (***PD SID***)
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	000b
B7…6	***Command Type***	01b (*Responder* ***ACK***)
B5	***Reserved***	0
B4…0	***Command***	2 (***Discover Identity***)
	ID Header VDO	
B31	***USB Communications Capable as USB Host***	0 (not *USB Communications* capable as a *USB Host*)
B30	***USB Communications Capable as a*** *USB Device*	1 (data capable as a Device)
B29…27	***SOP' Product Type (Cable Plug/VPD)***	001b (*Hub*)
B26	***Modal Operation Supported***	0 (doesn't support Modes)
B25…16	***Reserved***. ***Shall*** be set to zero.	0
B15…0	16-bit unsigned integer. ***USB Vendor ID***	USB-IF assigned *VID* for this *Hub* vendor
	Cert Stat VDO	
B31…0	32-bit unsigned integer	USB-IF assigned XID for this *Hub*
	Product VDO	
B31…16	16-bit unsigned integer. USB Product ID	Product ID assigned by the *Hub* vendor
B15…0	16-bit unsigned integer. bcdDevice	Device version assigned by the *Hub* vendor

C.2 Discover SVIDs Example

C.2.1 Discover SVIDs Command request

Table C.4, "Discover SVIDs Command request from Initiator Example" below shows the contents of the key fields in the *Message Header* and *VDM Header* for an *Initiator* sending a ***Discover SVIDs*** *Command* request.

Table C.4 Discover SVIDs Command request from Initiator Example

Bit(s)	Field	Value
Message Header		
15	***Reserved***	0
14…12	***Number of Data Objects***	1 (*VDM Header*)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
VDM Header		
B31…16	***Standard or Vendor ID (SVID)***	0xFF00 (***PD SID***)
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	000b
B7…6	***Command Type***	00b (*Initiator*)
B5	***Reserved***	0
B4…0	***Command***	2 (***Discover SVIDs***)

C.2.2 Discover SVIDs Command response

Table C.5, "Discover SVIDs Command response from Responder Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for a *Responder* returning VDOs in response to a ***Discover SVIDs*** *Command* request. In this illustration, the value 3 in the Message Header ***Number of Data Objects*** field indicates that one VDO containing the supported SVIDs would be returned followed by a terminating VDO.

Note: The last VDO returned (the terminator of the transmission) contains zero value SVIDs. If a SVID value is zero, it is not used.

Table C.5 Discover SVIDs Command response from Responder Example

Bit(s)	Field	Value
	Message Header	
15	***Reserved***	0
14…12	***Number of Data Objects***	3 (*VDM Header* + 2*VDO)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
	VDM Header	
B31…16	***Standard or Vendor ID (SVID)***	0xFF00 (***PD SID***)
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	000b
B7…6	***Command Type***	01b (*Responder* ***ACK***)
B5	***Reserved***	0
B4…0	***Command***	2 (***Discover SVIDs***)
	VDO 1	
B31…16	SVID 0	SVID value
B15…0	SVID 1	SVID value
	VDO 2	
B31…16	SVID 2	0x0000
B15…0	SVID 3	0x0000

C.3 Discover Modes Example

C.3.1 Discover Modes Command request

Table C.6, "Discover Modes Command request from Initiator Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for an *Initiator* sending a ***Discover Modes*** *Command* request. The *Initiator* of the ***Discover Modes*** *Command AMS* sends a Message Header with the ***Number of Data Objects*** field set to 1 followed by a *VDM Header* with the ***Command Type*** (B7…6) set to zero indicating the *Command* is from an *Initiator* and the ***Command*** (B4…0) is set to 3 indicating Mode discovery.

Table C.6 Discover Modes Command request from Initiator Example

Bit(s)	Field	Value
Message Header		
15	***Reserved***	0
14…12	***Number of Data Objects***	1 (*VDM Header*)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
VDM Header		
B31…16	***Standard or Vendor ID (SVID)***	SVID for which Modes are being requested
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	000b
B7…6	***Command Type***	00b (*Initiator*)
B5	***Reserved***	0
B4…0	***Command***	3 (***Discover Modes***)

C.3.2 Discover Modes Command response

The *Responder* to the ***Discover Modes*** *Command* request returns a Message Header with the ***Number of Data Objects*** field set to a value of 1 to 7 (the actual value is the number of Mode objects plus one) followed by a *VDM Header* with the ***Command Type*** (B7…6) set to 1 indicating the *Command* is from a *Responder* and the ***Command*** (B4…0) set to 3 indicating the following objects describe the Modes the device supports. If the ID is a *VID*, the structure and content of the VDO is left to the vendor. If the ID is a SID, the structure and content of the VDO is defined by the Standard.

Table C.7, "Discover Modes Command response from Responder Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for a *Responder* returning VDOs in response to a ***Discover Modes*** *Command* request. In this illustration, the value 2 in the Message Header ***Number of Data Objects*** field indicates that the device is returning one VDO describing the Mode it supports. It is possible for a *Responder* to report up to six different Modes.

Table C.7 Discover Modes Command response from Responder Example

Bit(s)	Field	Value
	Message Header	
15	***Reserved***	0
14…12	***Number of Data Objects***	2 (*VDM Header* + 1 Mode VDO)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
	VDM Header	
B31…16	***Standard or Vendor ID (SVID)***	SVID for which Modes were requested
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	000b
B7…6	***Command Type***	01b (*Responder **ACK***)
B5	***Reserved***	0
B4…0	***Command***	3 (***Discover Modes***)
	Mode VDO	
B31…0	Mode 1	Standard or Vendor defined Mode value

C.4 Enter Mode Example

C.4.1 Enter Mode Command request

The *Initiator* of the ***Enter Mode*** *Command* request sends a Message Header with the ***Number of Data Objects*** field set to 1 followed by a *VDM Header* with the ***Command Type*** (B7…6) set to 0 indicating the *Command* is from an *Initiator* and the ***Command*** (B4…0) set to 4 to request the *Responder* to enter its mode of operation and sets the Object Position field to the desired functional VDO based on its offset as received during Discovery.

Table C.8, "Enter Mode Command request from Initiator Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for an *Initiator* sending an ***Enter Mode*** *Command* request.

Table C.8 Enter Mode Command request from Initiator Example

Bit(s)	Field	Value
	Message Header	
15	***Reserved***	0
14…12	***Number of Data Objects***	1 (*VDM Header*)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
	VDM Header	
B31…16	***Standard or Vendor ID (SVID)***	SVID for the Mode being entered
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	001b (a one in this field indicates a request to enter the first Mode in list returned by ***Discover Modes***)
B7…6	***Command Type***	00b (*Initiator*)
B5	***Reserved***	0
B4…0	***Command***	4 (***Enter Mode***)

C.4.2 Enter Mode Command response

The *Responder* that is the target of the ***Enter Mode*** *Command* request sends a Message Header with the ***Number of Data Objects*** field set to a value of 1 followed by a *VDM Header* with the ***Command Type*** (B7…6) set to 1 indicating the *Command* is from a *Responder* and the ***Command*** (B4…0) set to 4 indicating the *Responder* has entered the Mode and is ready to operate.

Table C.9, "Enter Mode Command response from Responder Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for a *Responder* sending an ***Enter Mode*** *Command* response with an ***ACK***.

Table C.9 Enter Mode Command response from Responder Example

Bit(s)	Field	Value
	Message Header	
15	***Reserved***	0
14…12	***Number of Data Objects***	1 (*VDM Header*)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
	VDM Header	
B31…16	***Standard or Vendor ID (SVID)***	SVID for the Mode entered
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	001b (offset of the Mode entered)
B7…6	***Command Type***	01b (*Responder* ***ACK***)
B5	***Reserved***	0
B4…0	***Command***	4 (***Enter Mode***)

C.4.3 Enter Mode Command request with additional VDO.

The *Initiator* of the ***Enter Mode*** *Command* request sends a Message Header with the ***Number of Data Objects*** field set to 2 indicating an additional VDO followed by a *VDM Header* with the ***Command Type*** (B7…6) set to zero indicating the *Command* is from an *Initiator* and the ***Command*** (B4…0) set to 4 to request the *Responder* to enter its mode of operation and sets the Object Position field to the desired functional VDO based on its offset as received during Discovery.

Table C.10, "Enter Mode Command request with additional VDO Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for an *Initiator* sending an ***Enter Mode*** *Command* request with an additional VDO.

Table C.10 Enter Mode Command request with additional VDO Example

Bit(s)	Field	Value
	Message Header	
15	***Reserved***	0
14…12	***Number of Data Objects***	1 (*VDM Header*)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
	VDM Header	
B31…16	***Standard or Vendor ID (SVID)***	SVID for the Mode being entered
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	001b (a one in this field indicates a request to enter the first Mode in list returned by ***Discover Modes***)
B7…6	***Command Type***	00b (*Initiator*)
B5	***Reserved***	0
B4…0	***Command***	4 (***Enter Mode***)
	Including ***Optional*** Mode specific VDO	
B31…0	Mode specific	

C.5 Exit Mode Example

C.5.1 Exit Mode Command request

The *Initiator* of the ***Exit Mode*** *Command* request sends a Message Header with the ***Number of Data Objects*** field set to 1 followed by a *VDM Header* with the ***Command Type*** (B7 …6) set to zero indicating the *Command* is from an *Initiator* and the ***Command*** (B4…0) set to 5 to request the *Responder* to exit its Mode of operation.

Table C.11, "Exit Mode Command request from Initiator Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for an *Initiator* sending an ***Exit Mode*** *Command* request.

Table C.11 Exit Mode Command request from Initiator Example

Bit(s)	Field	Value
Message Header		
15	***Reserved***	0
14…12	***Number of Data Objects***	1 (*VDM Header*)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
VDM Header		
B31…16	***Standard or Vendor ID (SVID)***	SVID for the Mode being exited
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	001b (identifies the previously entered Mode by its Object Position that is to be exited)
B7…6	***Command Type***	00b (*Initiator*)
B5	***Reserved***	0
B4…0	***Command***	5 (***Exit Mode***)

C.5.2 Exit Mode Command response

The *Responder* that receives the ***Exit Mode*** *Command* request sends a Message Header with the ***Number of Data Objects*** field set to a value of 1 followed by a *VDM Header* with the ***Command Type*** (B7…6) set to 1 indicating the *Command* is from a *Responder* and the ***Command*** (B4…0) set to 5 indicating the *Responder* has exited the Mode and has returned to normal USB operation.

Table C.12, "Exit Mode Command response from Responder Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for a *Responder* sending an ***Exit Mode*** *Command* ***ACK*** response.

Table C.12 Exit Mode Command response from Responder Example

Bit(s)	Field	Value
	Message Header	
15	***Reserved***	0
14…12	***Number of Data Objects***	1 (*VDM Header*)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
	VDM Header	
B31…16	***Standard or Vendor ID (SVID)***	SVID for the Mode exited
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	001b (offset of the Mode to be exited)
B7…6	***Command Type***	01b (*Responder* ***ACK***)
B5	***Reserved***	0
B4…0	***Command***	5 (***Exit Mode***)

C.6 Attention Example

C.6.1 Attention Command request

The *Initiator* of the ***Attention*** *Command* request sends a Message Header with the ***Number of Data Objects*** field set to 1 followed by a *VDM Header* with the ***Command Type*** (B7 …6) set to zero indicating the *Command* is from an *Initiator* and the ***Command*** (B4…0) set to 6 to request attention from the *Responder*.

Table C.13, "Attention Command request from Initiator Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for an *Initiator* sending an ***Attention*** *Command* request.

Table C.13 Attention Command request from Initiator Example

Bit(s)	Field	Value
Message Header		
15	***Reserved***	0
14…12	***Number of Data Objects***	1 (*VDM Header*)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
VDM Header		
B31…16	***Standard or Vendor ID (SVID)***	SVID for which attention is being requested
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	001b (offset of the Mode requesting attention)
B7…6	***Command Type***	00b (*Initiator*)
B5	***Reserved***	0
B4…0	***Command***	6 (***Attention***)

C.6.2 Attention Command request with additional VDO.

The *Initiator* of the ***Attention*** *Command* request sends a Message Header with the ***Number of Data Objects*** field set to 2 indicating an additional VDO followed by a *VDM Header* with the ***Command Type*** (B7…6) set to zero indicating the *Command* is from a *Responder* and the ***Command*** (B4…0) set to 6 to request attention from the *Responder*.

Table C.14, "Attention Command request from Initiator with additional VDO Example" shows the contents of the key fields in the *Message Header* and *VDM Header* for an *Initiator* sending an ***Attention*** *Command* request with an additional VDO.

Table C.14 Attention Command request from Initiator with additional VDO Example

Bit(s)	Field	Value
Message Header		
15	***Reserved***	0
14…12	***Number of Data Objects***	2 (*VDM Header* + VDO)
11…9	***MessageID***	0…7
8	***Port Power Role***	0 or 1
7…6	***Specification Revision***	10b (*Revision 3.x*)
5…4	***Reserved***	0
3…0	***Message Type***	1111b (*Vendor Defined Message*)
VDM Header		
B31…16	***Standard or Vendor ID (SVID)***	SVID for which attention is being requested
B15	***VDM Type***	1 (*Structured VDM*)
B14…13	***Structured VDM Version (Major)***	01b (Version 2.0)
B12…11	***Structured VDM Version (Minor)***	01b (Version 2.1)
B10…8	***Object Position***	001b (offset of the Mode requesting attention)
B7…6	***Command Type***	000b (*Initiator*)
B5	***Reserved***	0
B4…0	***Command***	6 (***Attention***)
Including *Optional* Mode specific VDO		
B31…0	Mode specific	